

# Microlocal Morse Theory and Truncated Microsupport

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## Preface

In these notes, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

## 1 Notations

We consider a commutative unital ring  $\mathbf{k}$  of finite global dimension (e.g.  $\mathbf{Z}$  or a field). Let  $X$  be a real analytic manifold. We denote the cotangent bundle of  $X$  by  $\pi: T^*X \rightarrow X$ . We denote by  $D^b(\mathbf{k}_X)$  the bounded derived category of sheaves of  $\mathbf{k}$ -modules on  $X$ .

## 2 Truncated Microsupport

**Definition 2.1.** Let  $F \in D^b(\mathbf{k}_X)$ . The closed conic subset  $SS_k(F)$  of  $T^*X$  is defined by:  $p \notin SS_k(F)$  if and only if there exists an open conic neighborhood  $U$  of  $p$  such that for any  $x \in \pi(U)$  and for any real-valued  $C^1$ -function  $\varphi$  defined on a neighborhood of  $x$  such that  $\varphi(x) = 0$ ,  $d\varphi(x) \in U$ , one has

$$(2.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

### 2.1 Functorial Properties

**Proposition 2.2.** Let  $f: X \rightarrow Y$  be a morphism of real analytic manifolds and let  $F \in D^b(\mathbf{k}_X)$  such that  $f$  is proper on  $\text{supp}(F)$ . Then for any  $k \in \mathbf{Z}$ ,

$$(2.2) \quad SS_k(Rf_*F) \subset f_\pi f_d^{-1}(SS_k(F)).$$

The equality holds in case  $f$  is a closed embedding.

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### 3 Lemmas from [KS90]

Consider a projective system of abelian groups  $(M_n, \rho_{n,p})_n$  indexed by  $\mathbf{N}$ . One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any  $n \in \mathbf{N}$ , the decreasing sequence  $(\rho_{n,p}(M_p))_p$  of subgroups of  $M_n$  is stationary.

For a projective system of abelian groups  $(M_n)_n$ , we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(3.1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(3.2) \quad E_\infty^\bullet: \cdots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \cdots.$$

Denote by

$$(3.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

**Proposition 3.1** ([KS90, Proposition 1.12.4]). Assume that for each  $j \in \mathbf{Z}$ , the system  $(M_n^j)_n$  satisfies the ML condition. Then

- (a) for each  $k \in \mathbf{Z}$ , the map  $\Phi_k$  is surjective,
- (b) if moreover, for a given  $i$  the system  $(H^{i-1}(E_n^\bullet))_n$  satisfies the ML condition, then  $\Phi_i$  is bijective.

The following is called Kashiwara's lemma.

**Proposition 3.2** ([KS90, Proposition 1.12.6]). Let  $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$  be a projective system of sets indexed by  $\mathbf{R}$ . Assume that for each  $s \in \mathbf{R}$ , the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps  $\rho_{s_0, s_1}$  ( $s_0 \leq s_1$ ) are injective (resp. surjective).

#### ML procedure for sheaves

**Proposition 3.3** ([KS90, Proposition 2.7.1]). Let  $X$  be a topological space and let  $F \in D^+(\mathbf{Z}_X)$ . Let  $(U_n)_{n \in \mathbf{N}}$  be an increasing sequence of open subsets of  $X$ ,  $(Z_n)_{n \in \mathbf{N}}$  a decreasing sequence of closed subsets of  $X$ . Set  $U := \bigcup_n U_n$ ,  $Z := \bigcap_n Z_n$ .

- (i) For any  $j$ , the natural map  $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$  is surjective.
- (ii) Assume that for a given  $j$ , the projective system  $(H_{Z_n}^{j-1}(U_n; F))_n$  satisfies the ML condition. Then  $\varphi_j$  is bijective.
- (iii) Let now  $(X_n)_{n \in \mathbf{N}}$  be an increasing family of subsets of  $X$  satisfying  $X = \bigcup_n X_n$  and  $X_n \subset \text{Int}(X_{n+1})$  for all  $n$ . Assume that for a given  $j$ , the projective system  $(H^{j-1}(X_n; F))_n$  satisfies the ML condition. Then the natural map  $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$  is bijective.

## 4 Results

In this section, let  $M$  be a manifold.<sup>1</sup>

**Theorem 4.1.** Let  $F$  in  $D^b(\mathbf{k}_M)$ , and  $\psi: M \rightarrow \mathbf{R}$  be a real-valued function of class  $C^1$ , proper on  $\text{supp}(F)$ . Let  $a < b$  be real numbers. We assume  $d\psi(x) \notin \text{SS}_k(F)$  for  $a \leq \psi(x) < b$ . For  $t \in \mathbf{R}$ , we set  $M_t := \psi^{-1}(] - \infty, t[)$ . Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each  $j < k$ .<sup>2</sup>

**Lemma 4.2.** Let  $-\infty < a < b < +\infty$  be real numbers and  $G \in D^b(\mathbf{R})$ . Let  $k$  be an integer. If

$$\left( H_{[t, +\infty[}^j(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(]-\infty, b[; G) \xrightarrow{\sim} H^j(]-\infty, a[; G)$$

for all  $j < k$ .<sup>3</sup>

**Proof.** Let  $t \in [a, b[$ . We will denote by  $I_t$  the open interval  $]-\infty, t[$  and denote by  $Z_t$  the closed interval  $]-\infty, t[ (= \overline{I_t})$ . By applying the functor  $(\cdot)_{\overline{I_t}}$  to the distinguished triangle

$$\text{R}\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}),$$

we have

$$(\text{R}\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}).$$

Applying the functor  $\text{R}\Gamma(\mathbf{R}; \cdot)$  to this, we have

$$(4.1) \quad (\text{R}\Gamma_{[t, +\infty[}(G))_t \rightarrow \text{R}\Gamma(Z_t; G) \rightarrow \text{R}\Gamma(I_t; G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}).$$

<sup>1</sup>Can  $M$  be of class  $C^\infty$ ?

<sup>2</sup>For  $j = k$ , this morphism might be injective.

<sup>3</sup>For  $k$ , this morphism might be injective. (not written in this draft.)

Then consider the following conditions for  $j \in \mathbf{Z}$ ,  $s \in [a, b[$ , and  $t \in ]a, b]$ :

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G), \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_t; G). \end{aligned}$$

By hypothesis, it follows from (4.1) that  $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$  for  $t \in [a, b[$ .<sup>4</sup> Since  $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$ , (a)<sub>s</sub><sup>j</sup> holds for any  $j < k$  and  $s \in [a, b[$ .<sup>5</sup>

Moreover, since we assume  $G$  is a bounded complex, (b)<sub>t</sub><sup>j</sup> holds for  $j \ll 0$  and for all  $t \in ]a, b]$ . Let us denote such  $j$  by  $j_0$ . Let us argue by induction on  $j$ . Assume (b)<sub>t</sub><sup>j</sup> holds for all  $t \in ]a, b[$  and all  $j \leq j_0$ . Applying Proposition 3.2, we find the isomorphisms

$$(4.2) \quad H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j \leq j_0 \text{ and all } a < s \leq t < b.$$

Let  $G \xrightarrow{\sim} G^\bullet$  be a flabby resolution. Let  $t \in ]a, b]$ . Consider a complex

$$E_n^\bullet = \Gamma(I_{t-\frac{1}{n}}; G^\bullet).$$

Since  $G^\bullet$  is flabby, we have  $E_n^j \cong E_\infty^j$  and the projective system  $(E_n^j)_n$  satisfies the ML condition for all  $j \in \mathbf{Z}$ .

By (4.2), the projective system  $(H^{j_0}(E_n^\bullet))_n$  satisfies the ML condition (because we have  $a < t - \frac{1}{n} < t$  for  $n \gg 0$ ). Applying Proposition 3.1 to  $(E_n^\bullet)_n$ , it follows that (b)<sub>t</sub><sup>j</sup> holds for  $j_0 + 1$  and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) & \xrightarrow{\sim} \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ & \cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ & \cong \varprojlim_{s<t} H^{j_0+1}(I_s; G). \end{aligned}$$

Again by Proposition 3.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leq t \leq b,$$

and injections

$$H^j(I_s; G) \hookrightarrow H^j(I_t; G)$$

for  $j = k$  and for all  $a < s \leq t \leq b$ .

Finally, using (4.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leq b.$$

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<sup>4</sup>Similarly, we have injections  $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$  for  $t \in [a, b[$ .

<sup>5</sup>Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all  $s \in [a, b[$ .

Similarly, for  $j = k$ , we have

$$H^k(I_b; G) \cong \varinjlim_{t > s} H^k(I_t; G) \hookrightarrow H^k(I_s; G) \hookrightarrow H^k(I_a; G).$$

□

**Proof of Theorem 4.1.** First, we have

$$H^j(M_t; F) \cong H^j \mathrm{R}\Gamma([- \infty, t[; \mathrm{R}\psi_* F)$$

for  $t \in I$ . Since we assume that  $d\psi(x) \notin \mathrm{SS}_k(F)$  for  $a \leq \psi(x) < b$ , and by (2.2), the formula  $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$  holds. Hence we have

$$(H^j \mathrm{R}\Gamma_{[\psi(x), +\infty[}(\mathrm{R}\psi_* F))_{\psi(x)} = 0$$

for each  $j \leq k$ . Then we can apply Lemma 4.2 and have

$$H^j([- \infty, b[; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j([- \infty, a[; \mathrm{R}\psi_* F)$$

for all  $j < k$ . Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

for all  $j < k$ .

□

## 5 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

**Proposition 5.1.** Let  $X$  be a Hausdorff space,  $F \in \mathcal{D}^b(\mathbf{Z}_X)$ , and let  $(U_t)_{t \in \mathbf{R}}$  be a family of open subsets of  $X$ . We assume the following conditions.

- (i)  $U_t = \bigcup_{s < t} U_s$ , for all  $t \in \mathbf{R}$ .
- (ii) For all pairs  $(s, t)$  with  $s \leq t$ , the set  $\overline{U_t - U_s} \cap \mathrm{supp} F$  is compact.
- (iii) Setting  $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$ , we have for all pairs  $(s, t)$  with  $s \leq t$ , and all  $x \in Z_s - U_t$ ,
$$(H_{X - U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

Then we have for all  $t \in \mathbf{R}$ , the isomorphisms

$$H^j\left(\bigcup_s U_s; F\right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k$$

and the injections

$$H^k\left(\bigcup_s U_s; F\right) \hookrightarrow H^k(U_t; F).$$

**Proof.** Consider the following conditions for  $j \in \mathbf{Z}$ ,  $s$ , and  $t$ :

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(U_s; F) \xleftarrow{\sim} H^j(U_t; F). \end{aligned}$$

Let us prove  $(a)_s^j$  for  $s \in \mathbf{R}$  and  $j < k$ . We can assume that  $\overline{U_t - U_s}$  is compact for any  $s \leq t$  by replacing  $X$  by  $\text{supp } F$ . Consider the distinguished triangle

$$(\mathbf{R}\Gamma_{X - U_t}(F))|_{Z_s} \rightarrow (\mathbf{R}\Gamma_{X - U_s}(F))|_{Z_s} \rightarrow (\mathbf{R}\Gamma_{U_t - U_s}(F))|_{Z_s} \xrightarrow{+1}.$$

By hypothesis (iii), it follows that

$$(H_{X - U_t}^j(F))|_{Z_s} = 0, \quad (H_{X - U_s}^j(F))|_{Z_s} = 0$$

for any  $j \leq k$  and then

$$(H_{U_t - U_s}^j(F))|_{Z_s} = 0 \quad \text{for any } j \leq k - 1.$$

Thus for any  $j < k$  and  $t \geq s$ , we have

$$\begin{aligned} (5.1) \quad 0 &= \Gamma(Z_s; H_{U_t - U_s}^j(F)) \\ (5.2) \quad &= \varinjlim_{U \supset Z_s} H^j(U \cap U_t; H) \end{aligned}$$

□

**Remark 5.2.** Lemma 4.2 follows from Proposition 5.1. Indeed, putting  $X = ]-\infty, b[$ ,  $F = G|_{I_b}$ , and  $U_t = I_t$ , we find that

- (i)  $\bigcup_{s < t} I_s = \bigcup_{s < t} ]-\infty, s[ = ]-\infty, t[ = I_t$  for all  $t \in \mathbf{R}$ .
- (ii) If  $s \leq t$ , then  $\overline{I_t - I_s} \cap \text{supp } G = \overline{[s, t[} \cap \text{supp } G = [s, t] \cap \text{supp } G$  is compact.
- (iii) For  $s \in \mathbf{R}$ , we have

$$Z_s = \bigcap_{t > s} \overline{I_t - I_s} = \bigcap_{t > s} \overline{[s, t[} = \bigcap_{t > s} [s, t] = \{s\},$$

and

$$Z_s - I_t = \{s\} - ]-\infty, t[ = \begin{cases} \{s\} & \text{if } s = t, \\ \emptyset & \text{if } s < t. \end{cases}$$

By hypothesis of Lemma 4.2, for all  $s \in \{s\}$ ,

$$H_{X - I_s}^j(F)_s = H_{[s, \infty[}^j(F)_s = 0.$$

Then we can apply Proposition 5.1, and we have the isomorphisms for all  $t \in [a, b[$  and  $j < k$ ,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k(I_b; F) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F)$$

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